Zero human in the inner loop

Position note on why Pathfinder exists

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1 The argument

Assume that AI agents built on the latest language models soon match or surpass humans at solving and checking most scientific and technical tasks, as is already the case in mathematics (Navier–Stokes). Then the scarce resource in research shifts from solving problems to formulating them, and idea generation is itself best done with these models. The human contribution therefore moves from producing ideas to designing the machines that produce them. Engineering agent systems that generate scientific ideas becomes the most important task in science, at least until agents can design such systems themselves.

Pathfinder (`https://github.com/vd1/pathfinder`) is a bet on this trend rather than a description of today: in its first campaign (nobel) its ideas held up better than its solutions, which still fall short of human papers on the same question. It runs with no human intervention, because human clocks are much slower than machine ones and any human step in the loop becomes its bottleneck.

2 Three qualifications

Zero intervention means the inner loop. No human acts during a run, but humans choose the corpus, the rater, its anchor scale and the prompts. In nobel, most of the real gains came from corrections at that level: reading sources as LaTeX instead of converted PDFs, validating the rater, adding a presentation pass. This is the "designing the machines" role of the argument, and it means human time still matters, in the outer loop, which is where Pathfinder has improved so far. The principle is therefore no human in the inner loop, not no human at all.

The bottleneck may be evaluation rather than generation. Even with strong solvers, a system that generates ideas needs a judge of which ideas matter. Today that judge is a language model rater anchored to human taste. A rater that tracks current consensus will find good ideas near the frontier but is unlikely to reward the ideas that move it. A large part of "agents engineering their own idea discovery systems" is building a judge that is not anchored to people.

Solving and generating may not separate cleanly. Nobel's weakest result was execution: about 2.5 points below the human papers realising the same idea, on the campaign's rating scale. Many good questions only appear while pushing on a hard problem, which is the intuition behind turning obstacles into subgoals in the growth solver. If generation and solving are coupled, generation cannot run far ahead of solving, and getting solving right remains part of the idea generation project rather than something to hand off.

3 Consequence for Pathfinder

None of the qualifications changes the conclusion. They change where to put the effort inside Pathfinder: the judge, and the path from solving back into question formation.